

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	19 FEB 2026
Team ID	LTVIP2026TMIDS65554
Project Name	Smart sorting: transfer Learning For Identifying Rotten Fruits and Vegetables
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form Registration through Gmail Registration through LinkedIN
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Image Upload	- Upload fruit/vegetable images via camera or file picker - Support for batch upload
FR-4	Image Classification	- Use Transfer Learning model to classify input as <i>Fresh</i> or <i>Rotten</i> - Display classification results with confidence score
FR-5	Dataset Management	- Admin panel for managing training images (add/edit/delete) - Option to retrain the model with updated dataset
FR-6	Results Dashboard	- View recent predictions - Filter results by date, category, and freshness status
FR-7	Feedback System	- Users can provide feedback on classification accuracy
FR-8	Report Generation	- Generate downloadable reports (CSV or PDF) of analysis results

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The interface should be user-friendly and intuitive for both tech-savvy and non-technical users such as farmers or shopkeepers.
NFR-2	Security	User data and uploaded images should be securely stored. Access to dataset management must be role-based.
NFR-3	Reliability	The system should perform consistently, ensuring accurate predictions under normal usage.
NFR-4	Performance	The model should return classification results within 2 seconds for a single image.
NFR-5	Availability	The system should be available 24/7 with minimum downtime, especially during peak usage hours.

NFR-6	Scalability	The system should support scaling for increased user load or larger datasets in the future.
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